

GROUP 7

K.DEEKSHITH

G.GOVARDHAN

T.CHETHAN

B.NARASIMHA

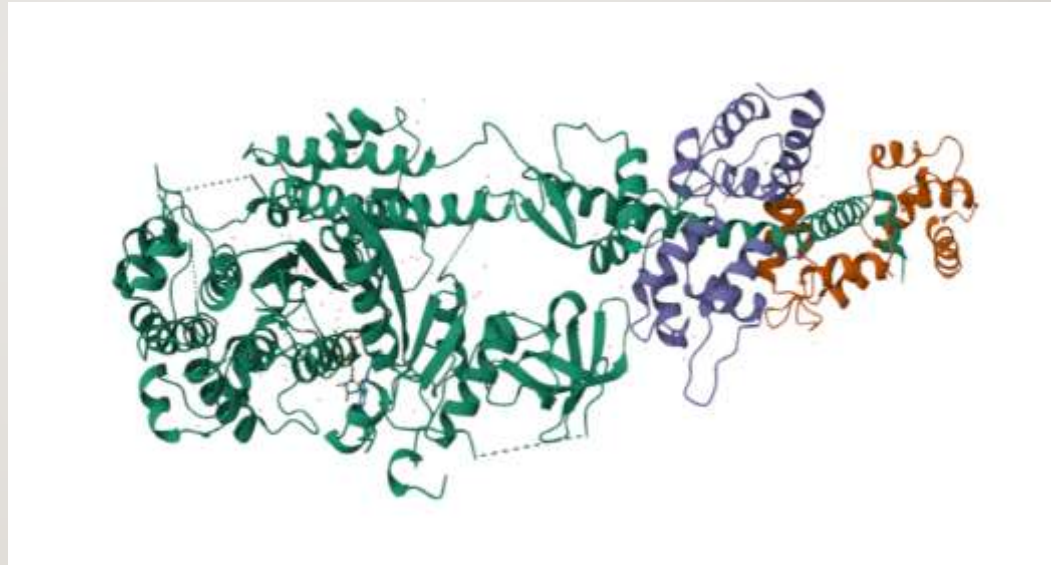
K.DHIRAJVARMA

E.VENKAT

THE PROTEIN WE HAVE CHOSEN

MYOSIN(1B7T)

ADENOSINE-5'-DIPHOSPHATE
 $\text{C}_{10} \text{H}_{15} \text{N}_5 \text{O}_{10} \text{P}_2$



ORGANISMS IN WHICH MYOSIN(IB7T) PROTEIN IS FOUND

IT IS MAINLY FOUND IN ARGOPECTEN IRRADIANS
COMMONLY KNOWN AS THE BAY SCALLOP OR
THE SEA SHELLS.



THE FUNCTION OF MYOSIN IN ARGOPECTEN IRRADIANS



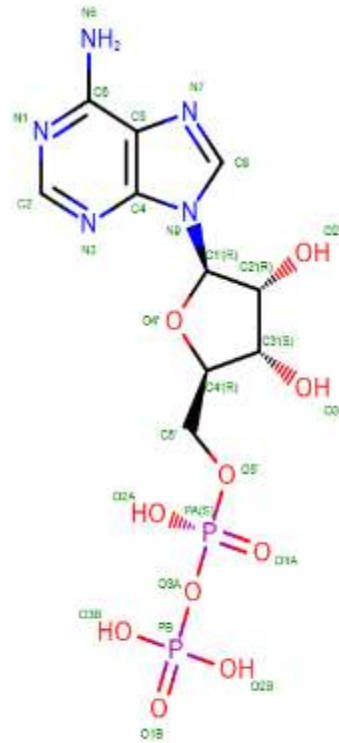
MYOSIN IS A TYPE OF MOTOR PROTEIN THAT INTERACTS WITH ACTIN FILAMENTS TO FACILITATE MUSCLE CONTRACTION IN VARIOUS ORGANISMS, INCLUDING MARINE ANIMALS LIKE THE BAY SCALLOP.



- 1) MUSCLE CONTRACTION AND MOVEMENT EFFICIENCY
- 2) ADAPTION TO THE MARINE ENVIRONMENT
- 3) MUSCLE FIBER TYPE SPECIALIZATION
- 4) STRUCTURAL AND FUNCTIONAL SPECIFICITY
- 5) EVOLUTIONARY INSIGHTS
- 6) IT HELPS IN LOCOMOTION



1B7T: Ligand ADP



LIGANDS

MOVIE SYNTAX

```
PyMOL>cmd.orient("ADP",animate=-1)
```

```
PyMOL>util.cba(33,"1B7T",_self=cmd)
```

```
Mset 1x1800
```

```
mview store,001
```

```
PyMOL>mview store,001
```

```
Turn y,180
```

```
Mview store,200
```

```
PyMOL>turn y,180
```

```
PyMOL>turn y,18
```

```
PyMOL>turn y,180
```

```
PyMOL>mview store,550
```

```
Mplay
```

```
mstop
```

```
PyMOL>fetch 1B7T
```



EXPLORING COLOR PALETTES WITH PALETTEER IN R

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A COMPREHENSIVE GUIDE TO BEAUTIFUL VISUALIZATIONS"



INTRODUCTION

WHAT IS PALETTER?"



- The paletter package in R provides a comprehensive collection of **color palettes** from the R ecosystem. It offers easy access to a broad range of color palettes from different packages, making it easier for users to choose appropriate color schemes for their visualizations.
- - Easy-to-use interface
- - Extensive collection of pre-built palettes
- - Customizable palette creation

PALETTEER FEATURES



- - Access to over 100 pre-built palettes
- - Palette search and filtering - Custom palette creation
- - Integration with popular R visualization packages (ggplot2, plotly)



INSTALLING AND LOADING



- Code snippet: `install.packages("paletteer")`
- `library(paletteer)`

- Outcomes:

```
The downloaded binary packages are in
  C:\Users\deeks\AppData\Local\Temp\RtmpWCTnhD\downloaded_packages
>
> install.packages("paletteer")
WARNING: Rtools is required to build R packages but is not currently installed. Please download and
install the appropriate version of Rtools before proceeding:

https://cran.rstudio.com/bin/windows/Rtools/
Installing package into 'C:/Users/deeks/AppData/Local/R/win-library/4.4'
(as 'lib' is unspecified)
trying URL 'https://cran.rstudio.com/bin/windows/contrib/4.4/paletteer_1.6.0.zip'
Content type 'application/zip' length 443634 bytes (433 KB)
downloaded 433 KB

package 'paletteer' successfully unpacked and MD5 sums checked

The downloaded binary packages are in
  C:\Users\deeks\AppData\Local\Temp\RtmpWCTnhD\downloaded_packages
> library(paletteer)
```

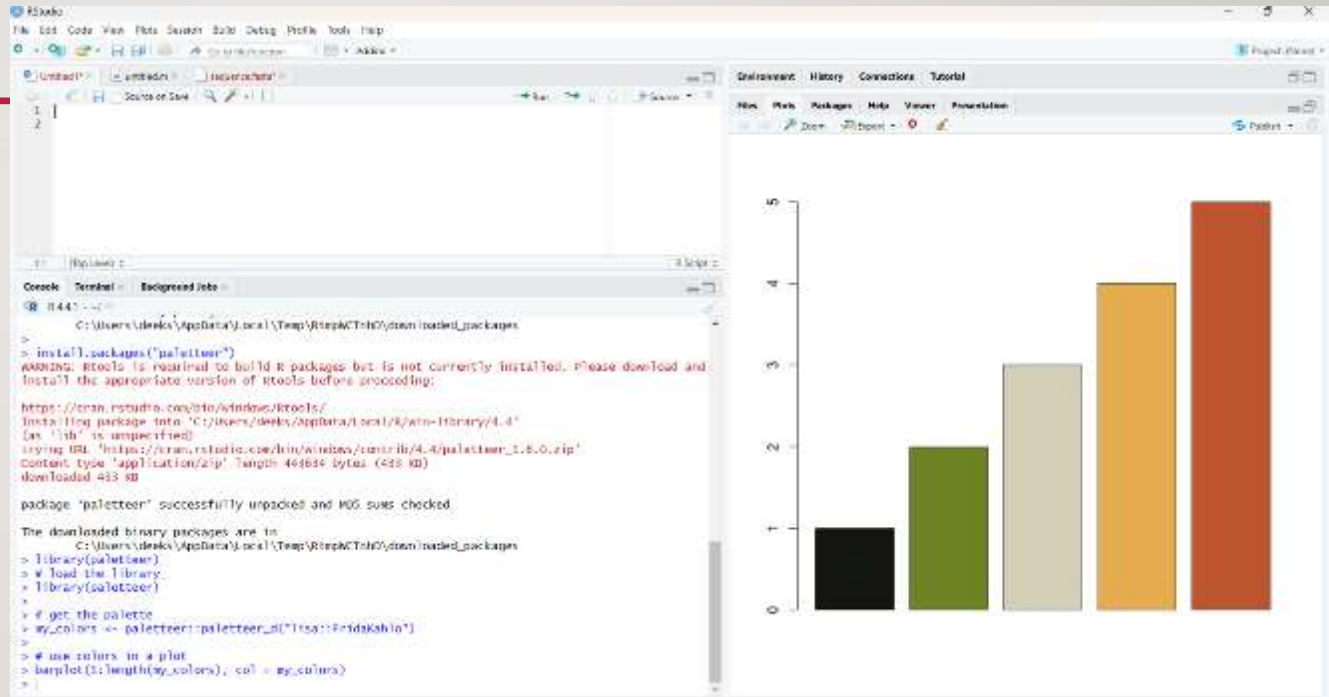
BASIC USAGE

- If you want to easily get started with `paletteer`, you can use the `paletteer_d()` function to access the palettes.
- `paletteer` has a straightforward syntax for accessing palettes:
`package_name::palette_name`.

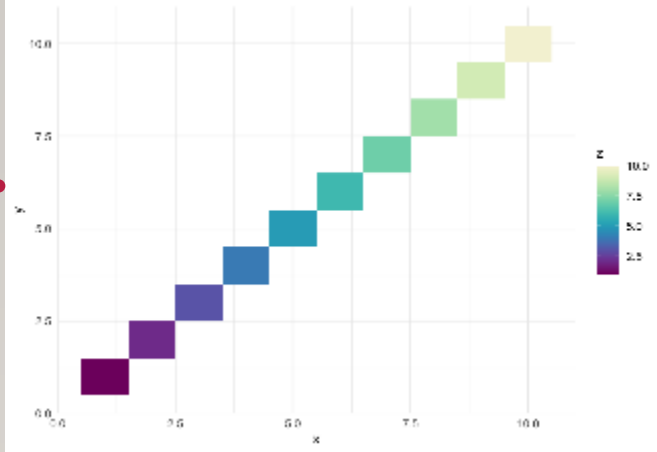
Here's a basic example with the `FridaKahlo` palette from the `lisa` package:

```
> # load the library
> library(paletteer)
>
> # get the palette
> my_colors <- paletteer::paletteer_d("lisa::FridaKahlo")
>
> # use colors in a plot
> barplot(1:length(my_colors), col = my_colors)
> |
```

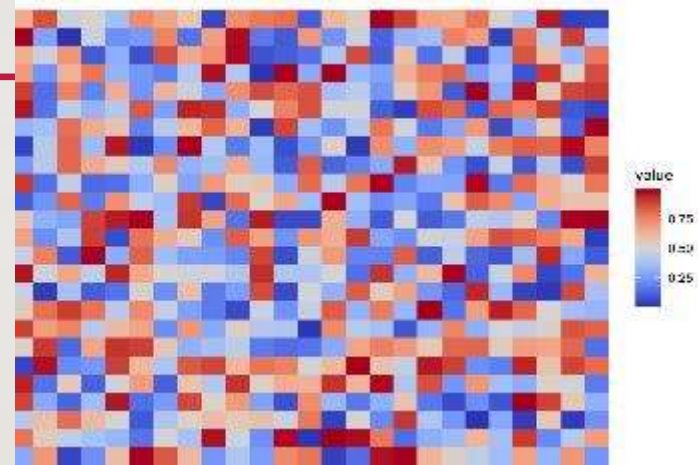
OUTPUT FOR THE CODE



THINGS TO DO WITH PALETTER

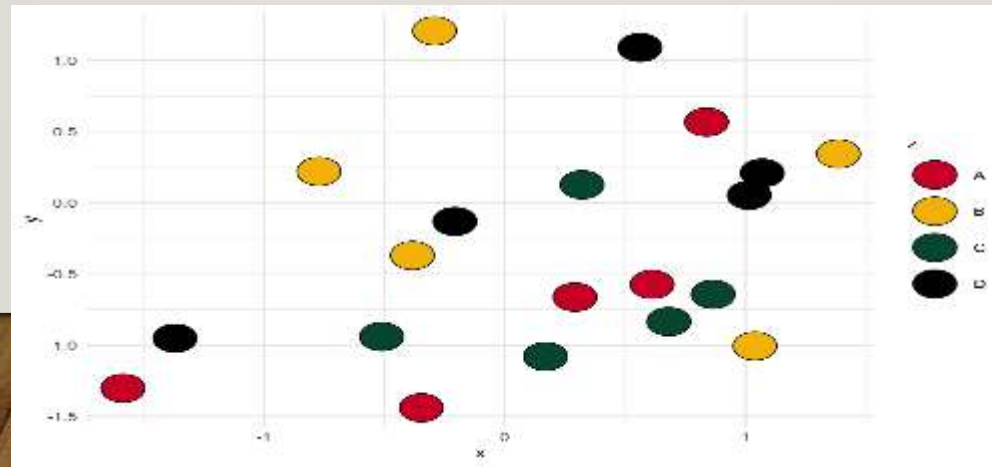


Sequential continuous palettes



Diverging continuous palettes

Qualitative discrete palettes

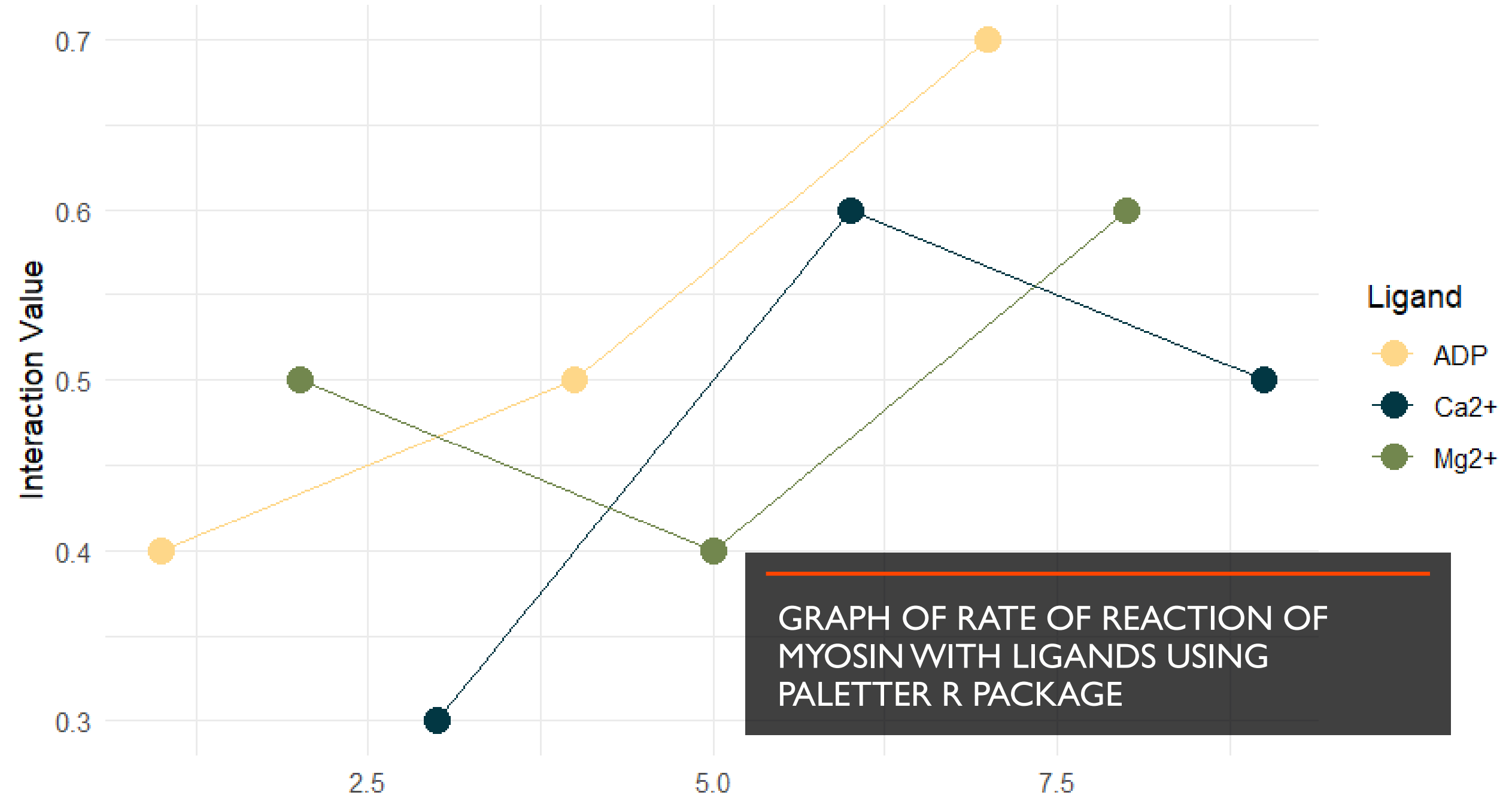


CONCLUSION SUMMARY

- The goal of **paletteer** is to be a comprehensive collection of color palettes in R using a common interface. Think of it as the “caret of palettes”.
- **Palettes have different types, and paletteer demands a specific function to access each type.**
- To colour the graphs .(simply).
- Call-to-action: "Explore paletteer today and enhance your visualizations!".



Myosin Interaction with ADP, Mg^{2+} , and Ca^{2+} vs Reaction Time



THANK YOU !



IT OFFERS MORE THAN **2000 COLOR PALETTES**

[HTTPS://R-GRAPH-GALLERY.COM/BEST-DATAVIZ-PACKAGES.HTML](https://r-graph-gallery.com/best-dataviz-packages.html)